

Laboratory work 3

We continue to work with the database from the previous laboratory works.

Take a full-page screenshot that covers the code and results of each task.

Tasks:

DML

1. Generated 200 rows in summary and inserted it.

The first screenshot shows a SQL query in pgAdmin 4 that inserts data into two tables: 'airline' and 'airport'. The query is as follows:

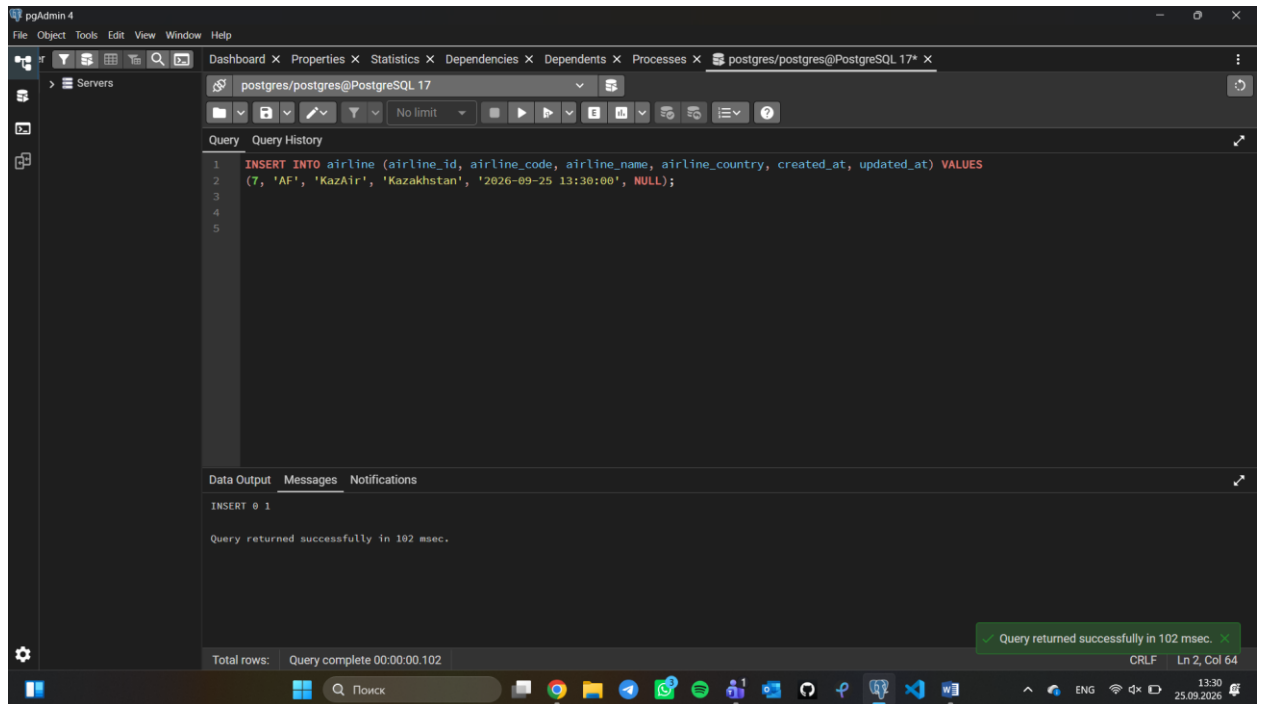
```
-- ВСТАВКА 200 СТРОК ДАННЫХ (DML)
--
-- 1. AIRLINE (6 строк)
INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at) VALUES
(1, 'AF', 'Air France', 'France', '2026-01-10 10:00:00', NULL),
(2, 'KC', 'Air Astana', 'Kazakhstan', '2026-01-10 10:05:00', NULL),
(3, 'DL', 'Delta Air Lines', 'USA', '2026-01-10 10:10:00', NULL),
(4, 'BA', 'British Airways', 'UK', '2026-01-10 10:15:00', NULL),
(5, 'LH', 'Lufthansa', 'Germany', '2026-01-10 10:20:00', NULL),
(6, 'EK', 'Emirates', 'UAE', '2026-01-10 10:25:00', NULL);
--
-- 2. AIRPORT (8 строк)
INSERT INTO airport (airport_id, airport_name, country, state, city, created_at, updated_at) VALUES
(1, 'CDG', 'France', 'Ile-de-France', 'Paris', '2026-01-11 08:00:00', NULL),
(2, 'PEK', 'China', 'Beijing', 'Beijing', '2026-01-11 08:05:00', NULL),
(3, 'ORY', 'France', 'Ile-de-France', 'Paris', '2026-01-11 08:10:00', NULL),
(4, 'AIA', 'Kazakhstan', 'Almaty', 'Almaty', '2026-01-11 08:15:00', NULL);
```

The second screenshot shows the results of a query that inserts 200 rows into a table named 'BOOKING'. The query is as follows:

```
SELECT * FROM BOOKING;
```

The results of the query are displayed in a table with the following columns: booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, and price. The table shows 20 rows of data, with the first row being (1, 1, 1, Website, 2026-04-01 09:00:00, [null], Confirmed, 500.00) and the last row being (20, 'Cleared', 20, '2026-05-21 09:05:00', NULL).

2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.



The screenshot shows the pgAdmin 4 interface. The 'Query' tab is active, displaying the following SQL query:

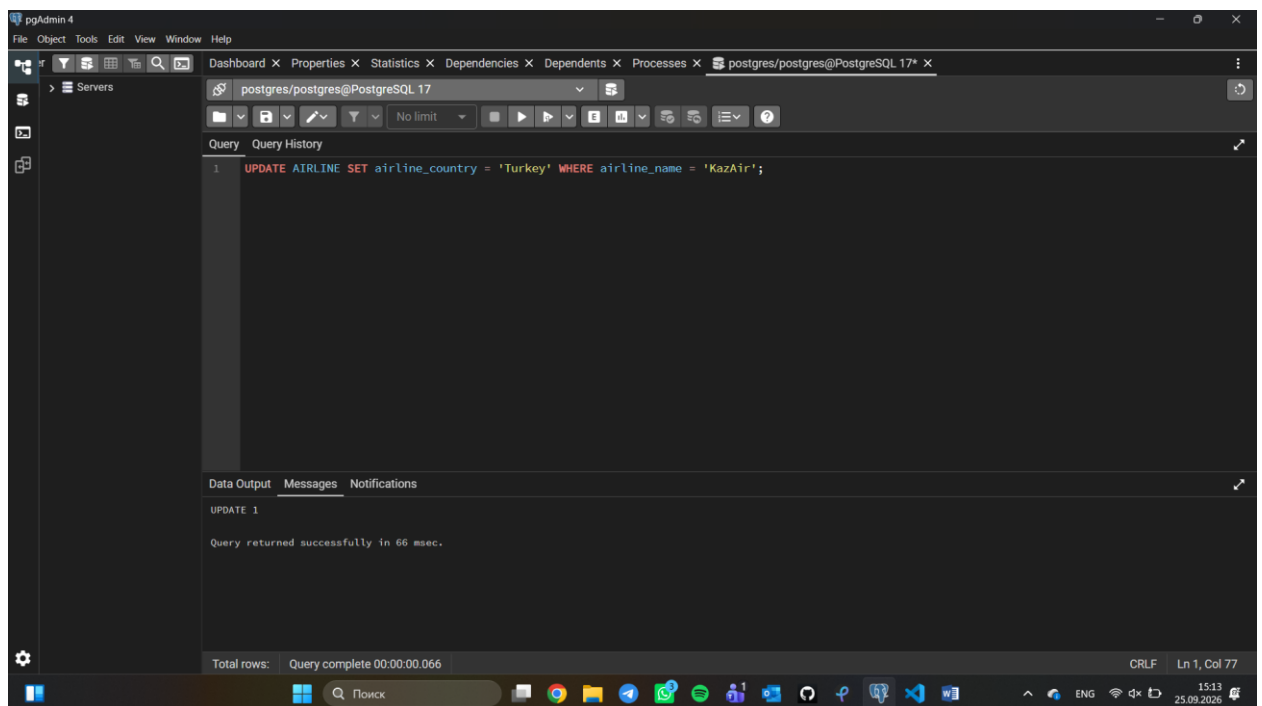
```
1 INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at) VALUES
2 (7, 'AF', 'KazAir', 'Kazakhstan', '2026-09-25 13:30:00', NULL);
3
4
5
```

The 'Data Output' tab shows the result of the query:

```
INSERT 0 1
Query returned successfully in 102 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 102 msec." The status bar also shows "Total rows: Query complete 00:00:00.102" and "Ln 2, Col 64".

3. Update the airline country "KazAir" to "Turkey".



The screenshot shows the pgAdmin 4 interface. The 'Query' tab is active, displaying the following SQL query:

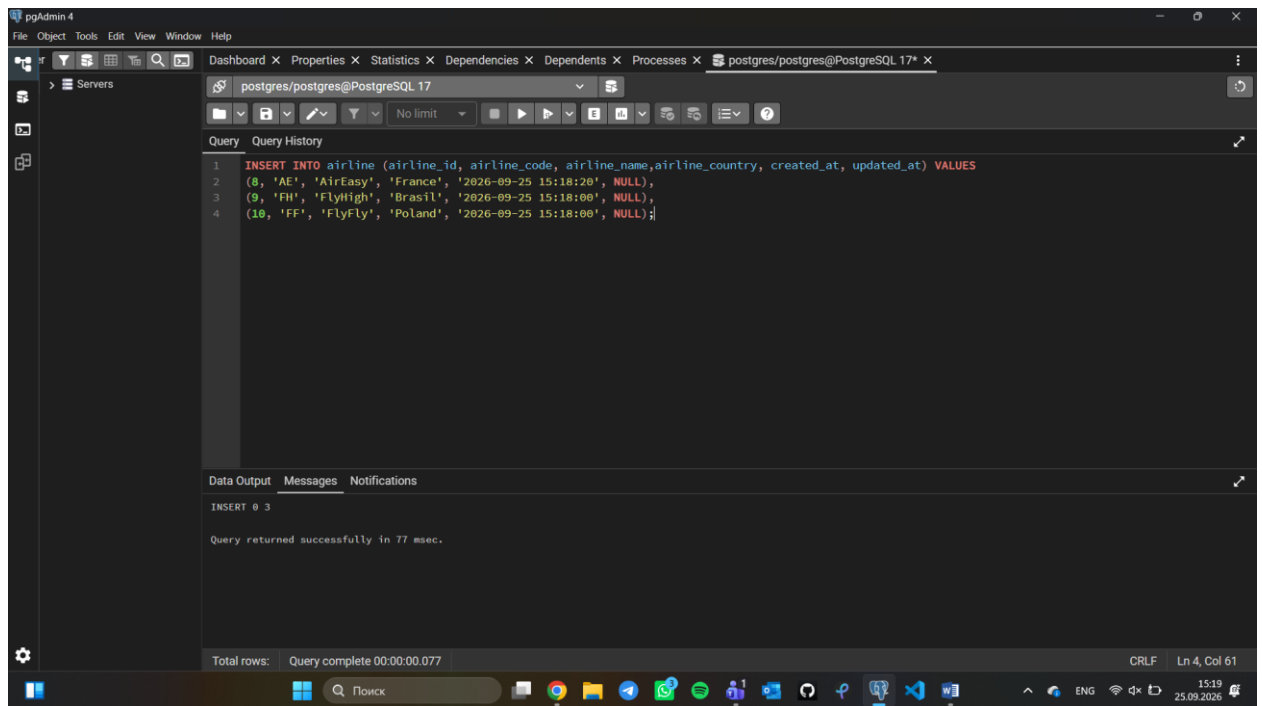
```
1 UPDATE AIRLINE SET airline_country = 'Turkey' WHERE airline_name = 'KazAir';
```

The 'Data Output' tab shows the result of the query:

```
UPDATE 1
Query returned successfully in 66 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 66 msec." The status bar also shows "Total rows: Query complete 00:00:00.066" and "Ln 1, Col 77".

4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".



The screenshot shows the pgAdmin 4 interface with a query window open. The query is an INSERT statement into the 'airline' table. The 'Data Output' tab shows the result of the query: 'INSERT 0 3' and 'Query returned successfully in 77 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.077' and 'Ln 4, Col 61'.

```
1 INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at) VALUES
2 (8, 'AE', 'AirEasy', 'France', '2026-09-25 15:18:20', NULL),
3 (9, 'FH', 'FlyHigh', 'Brasil', '2026-09-25 15:18:00', NULL),
4 (10, 'FF', 'FlyFly', 'Poland', '2026-09-25 15:18:00', NULL);
```

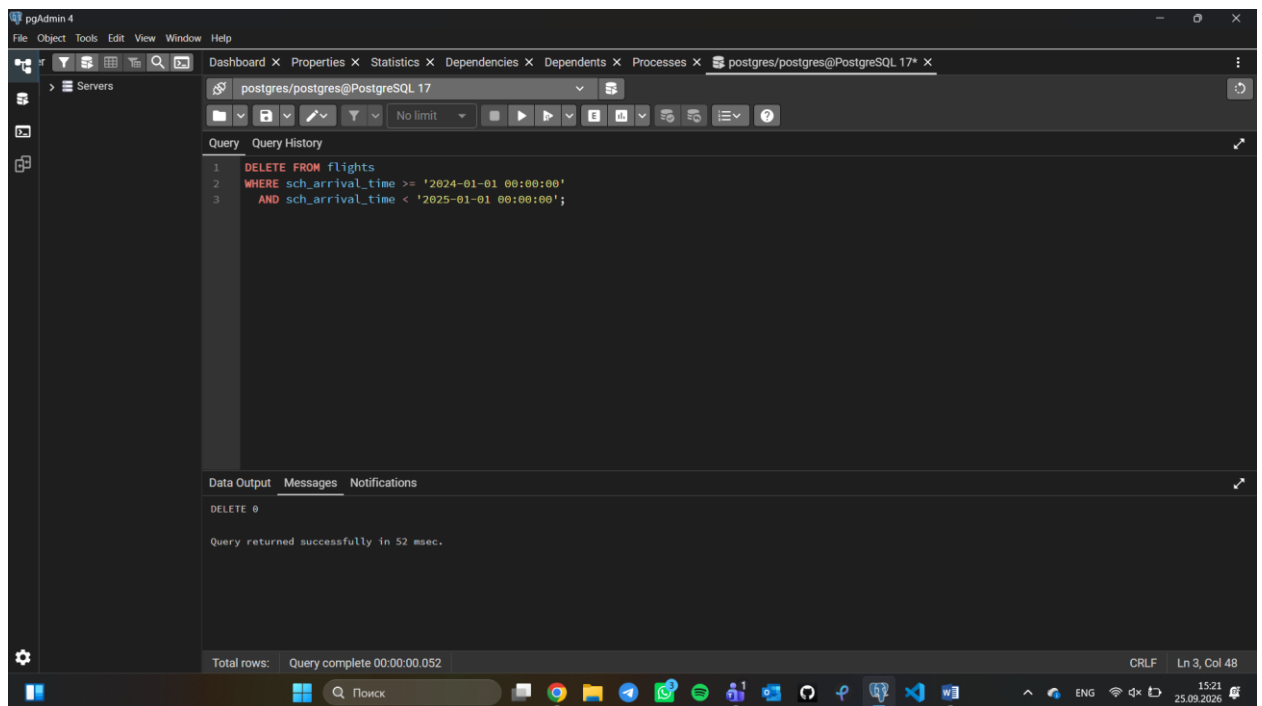
Data Output Messages Notifications

INSERT 0 3

Query returned successfully in 77 msec.

Total rows: Query complete 00:00:00.077 CRLF Ln 4, Col 61

5. Delete all flights whose arrival in 2024 year.



The screenshot shows the pgAdmin 4 interface with a query window open. The query is a DELETE statement from the 'flights' table. The 'Data Output' tab shows the result of the query: 'DELETE 0' and 'Query returned successfully in 52 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.052' and 'Ln 3, Col 48'.

```
1 DELETE FROM flights
2 WHERE sch_arrival_time >= '2024-01-01 00:00:00'
3 AND sch_arrival_time < '2025-01-01 00:00:00';
```

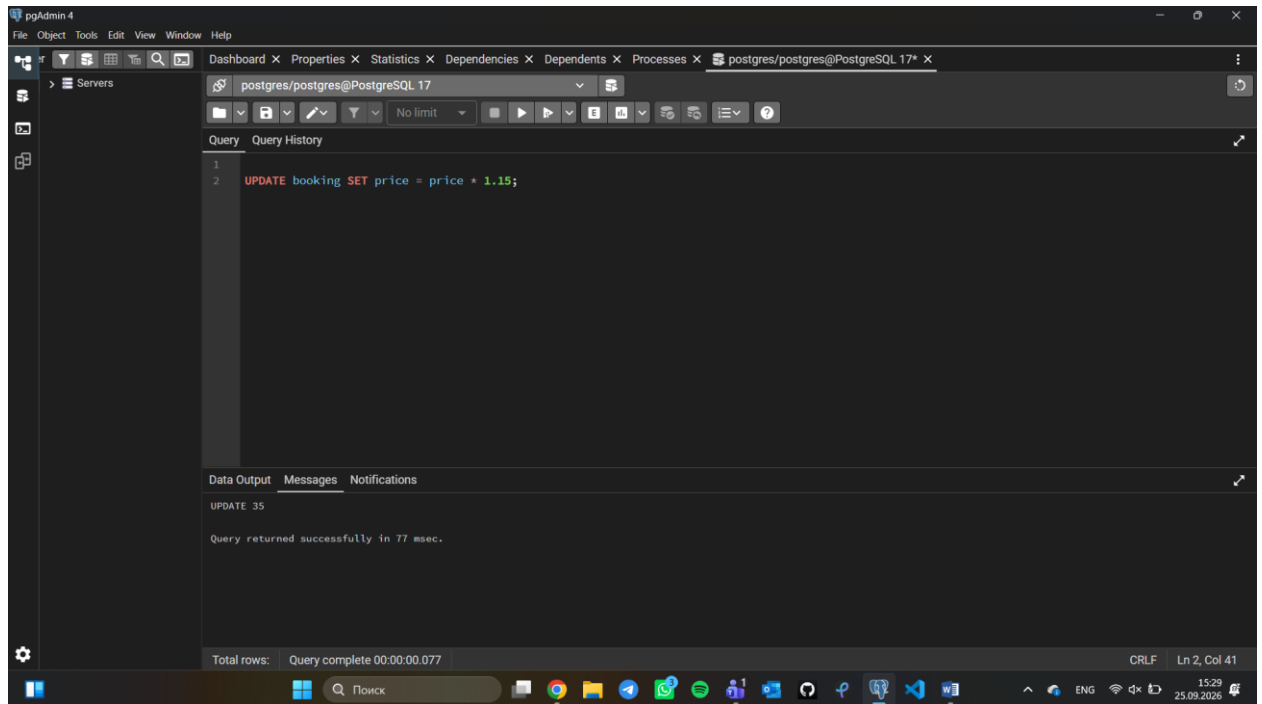
Data Output Messages Notifications

DELETE 0

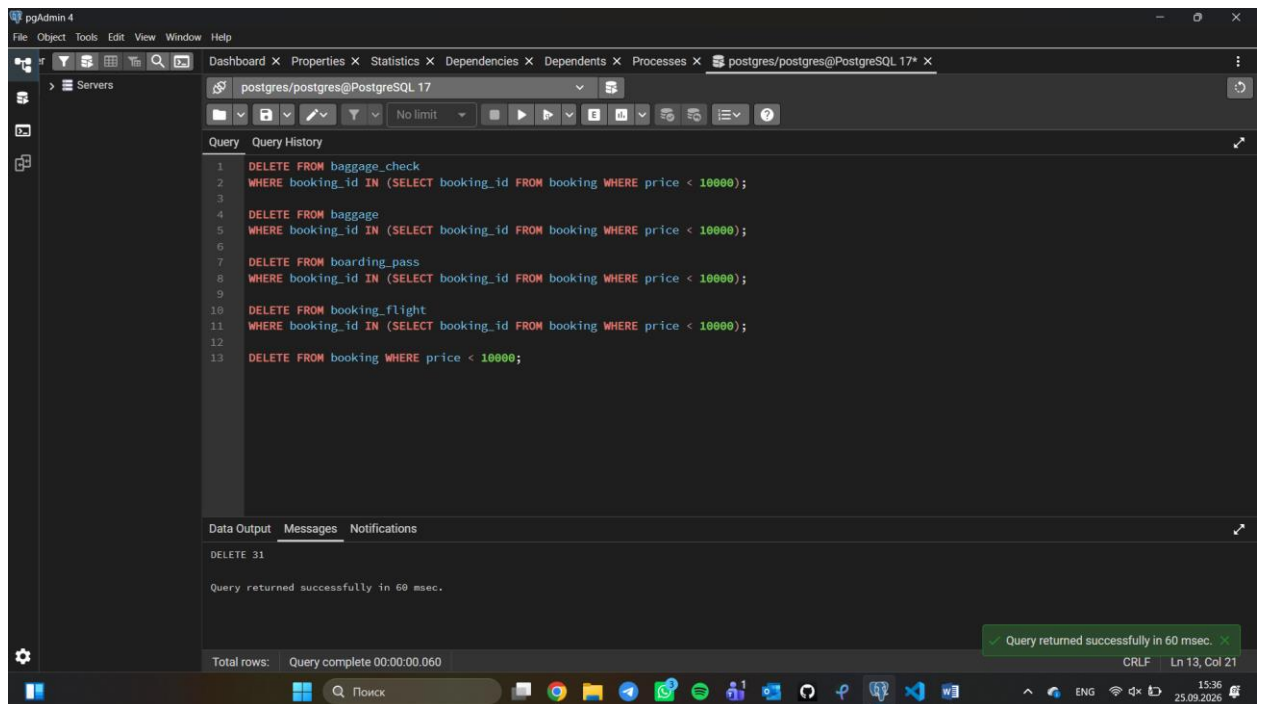
Query returned successfully in 52 msec.

Total rows: Query complete 00:00:00.052 CRLF Ln 3, Col 48

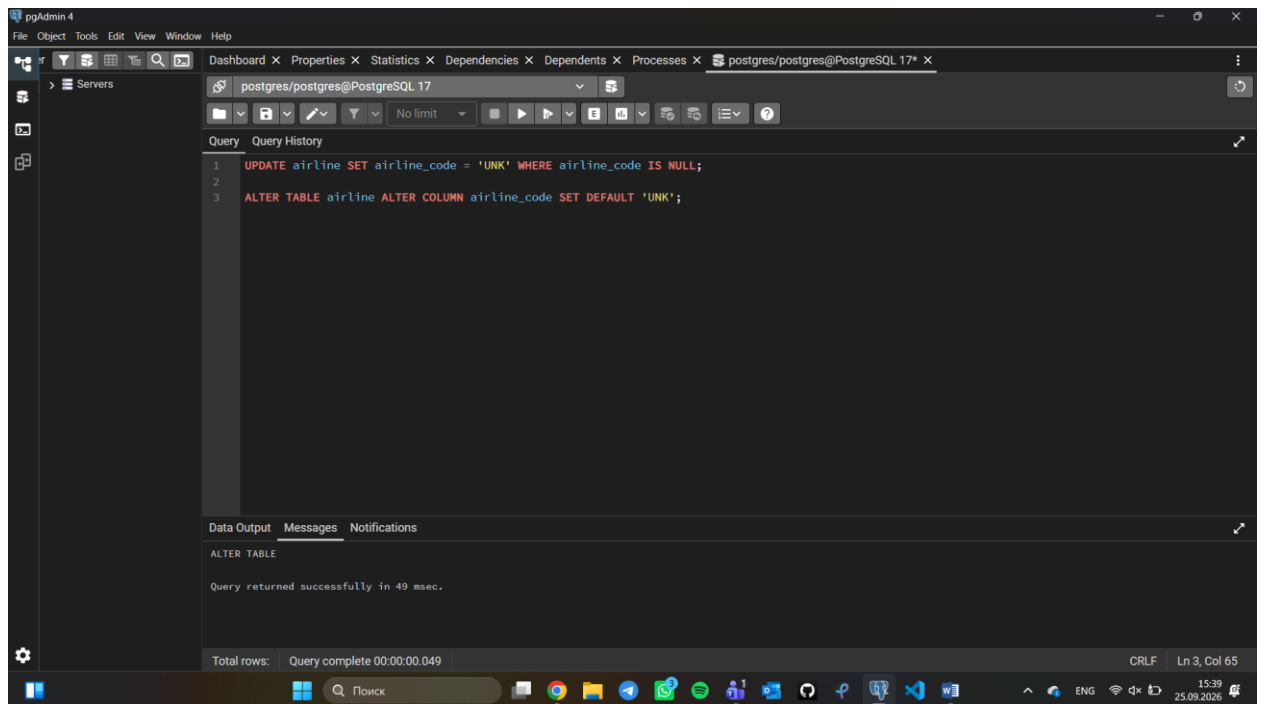
6. Increase the price of all tickets in booking table for flights by 15%.



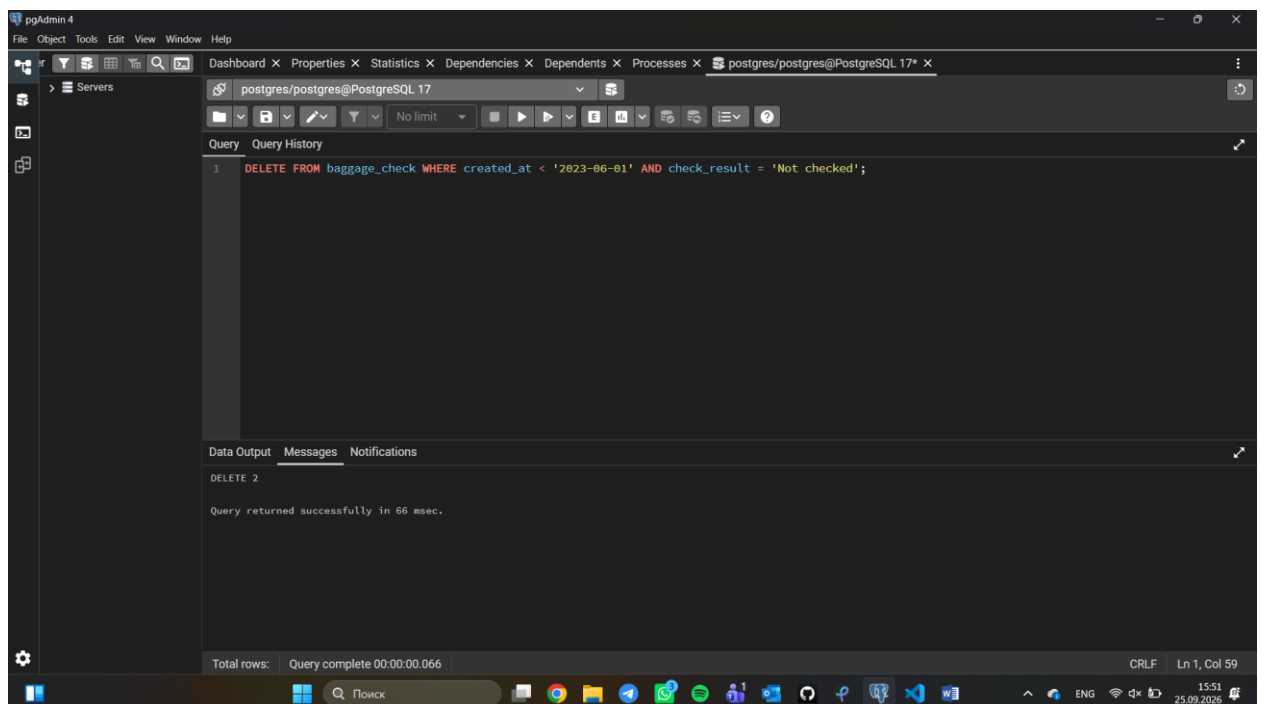
7. Delete all tickets where price is less than 10000.



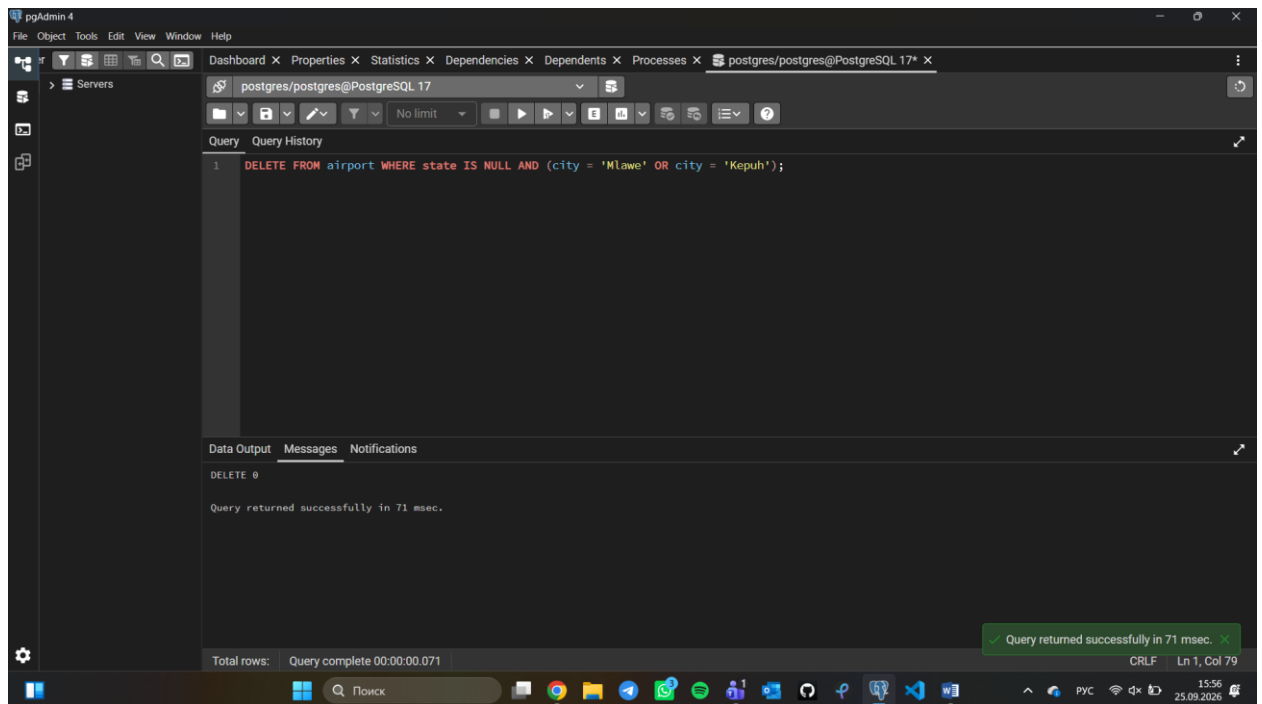
- Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).



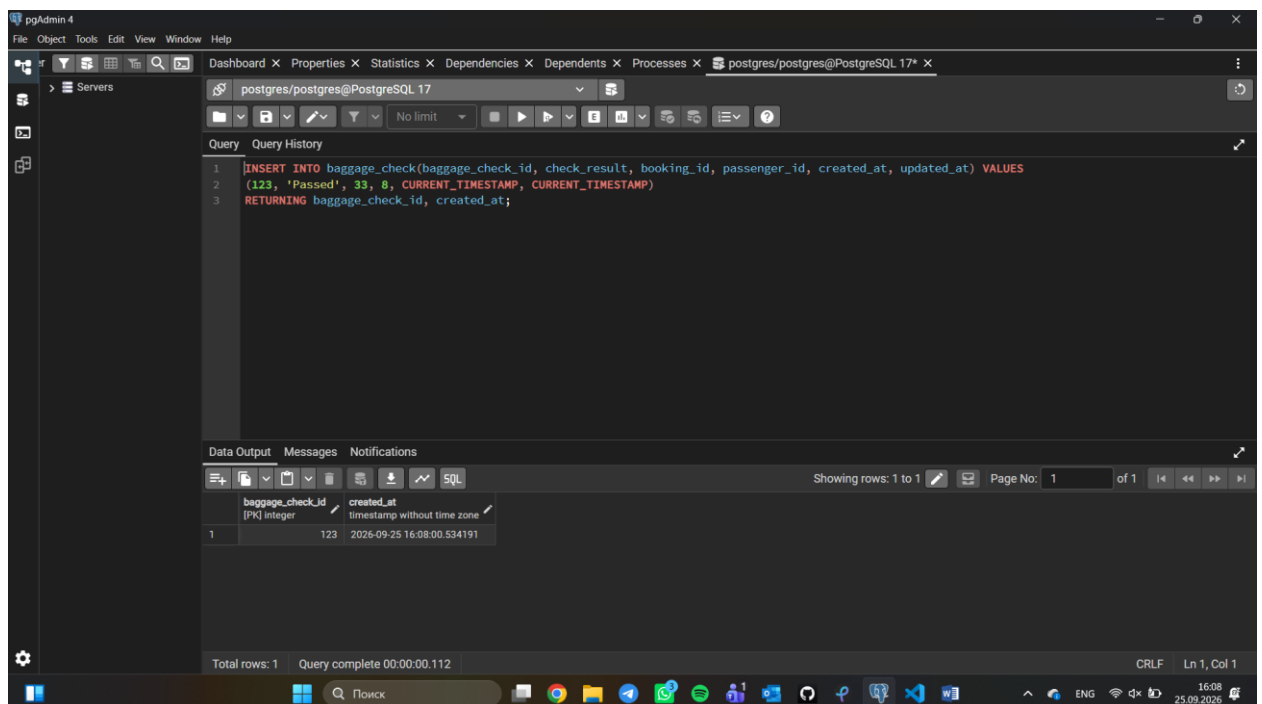
- Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.



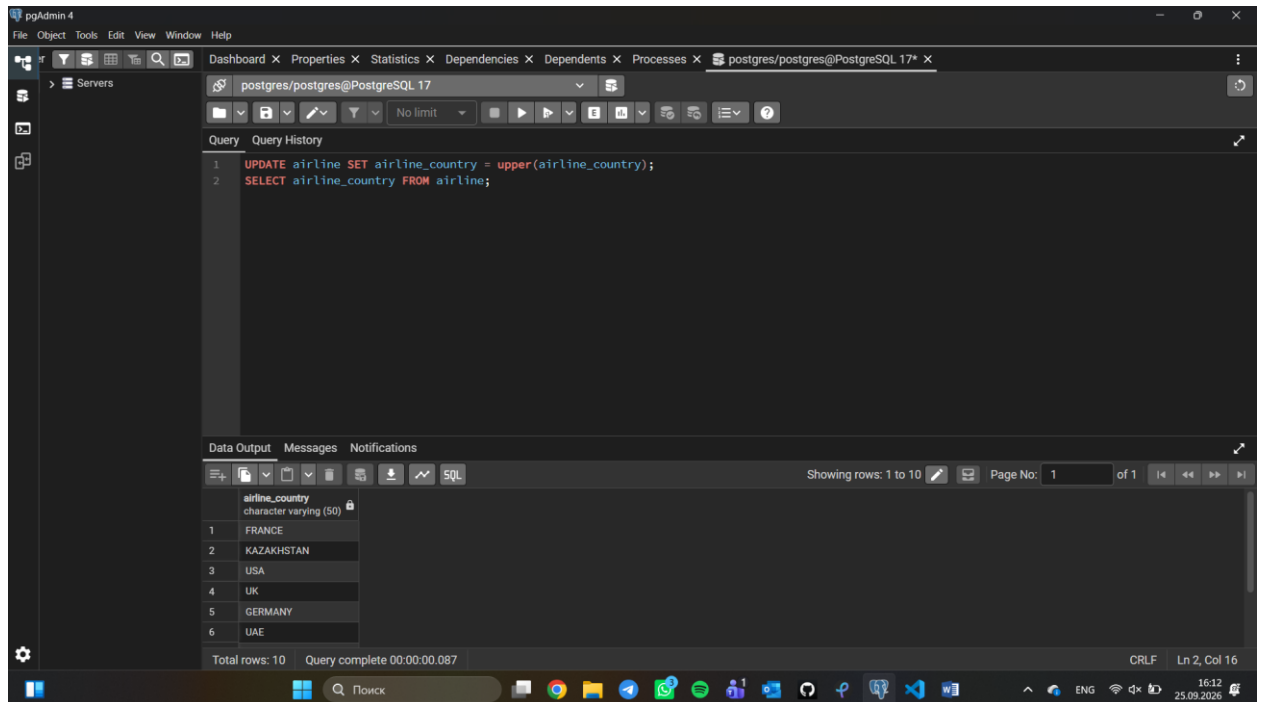
10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.



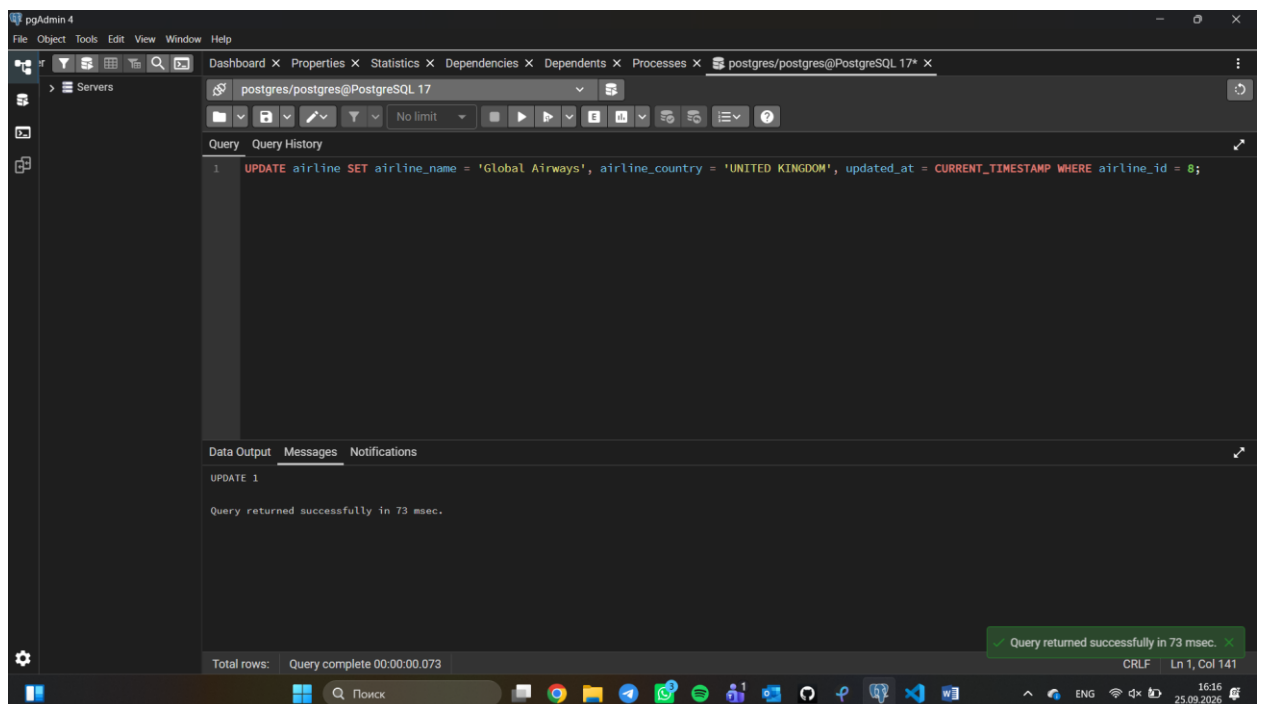
11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.



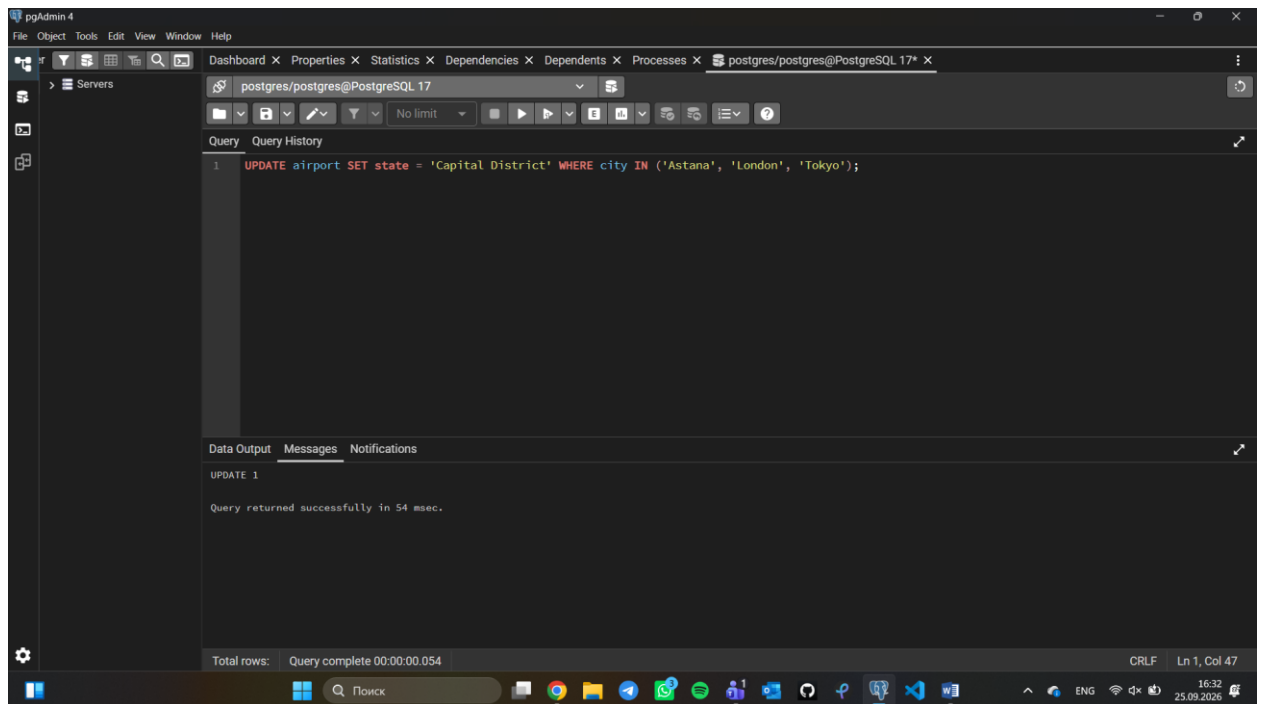
12. Convert all country names in the airline table to uppercase to ensure data consistency.



13. Update both the name and country for the airline with airline_id = 5. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating update_at to the current date.



14. Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'



15. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

